5. Discussion

5.1. Grain-size distribution

The grain-size data analysed in this study were originally presented by Wacha et al. (2021) and interpreted using end-member modelling (EMM). Here, the same dataset is reanalysed using CoDa. The two approaches address complementary aspects of the data. EMM partitions grain-size distributions into subpopulations that reflect sediment sources and transport processes, whereas the CoDa framework quantifies the magnitude and statistical significance of compositional changes between adjacent OIS stages and identifies the most stable grain-size fractions across climatic transitions. Rather than being redundant, the two methods provide complementary perspectives and enable cross-validation of the paleoenvironmental signal preserved in the Zmajevac LPS.

The dataset centres (Table 1) indicate that fine to coarse silt (16-44 µm) is the dominant fraction across all OIS groups, consistent with typical loess grain-size distributions in the Danube region (Novothny et al., 2011; Jipa, 2014; Vári et al., 2024). The most striking difference occurs between the OIS 4 and OIS 5 sample groups. OIS 5 is characterised by higher proportions of clay (< 5.5 µm) and very fine to fine silt (5.5-16 µm), and lower proportions of coarse silt (44-63 µm) and very fine and fine sand (63-200 µm), whereas OIS 4 shows the opposite pattern. This contrast aligns with Wacha et al. (2021), who associated EM1 (fine silt and clay) with pedogenic intensity and EM4 (very fine and fine sand) with glacial conditions.

Medium silt (16-44 µm) exhibits the lowest variation in log ratios and, consequently, the smallest clr variance (Table 2), indicating that its relative proportion remained effectively constant during environmental changes. In Wacha et al. (2021), this size range corresponds to EM2 and EM3, which are interpreted as products of longer-distance aeolian transport. From a process perspective, finer particles remain airborne longer and travel further, whereas coarser particles require stronger winds and are transported primarily by saltation. Medium silt appears to occupy an intermediate transport regime, insufficiently light to remain in prolonged suspension but readily mobilised under a wide range of wind conditions. This likely resulted in a relatively stable depositional flux, largely decoupled from glacial-interglacial variability.

In contrast, the largest variation in log ratios occurs between clay and coarse silt, clay and very fine and fine sand, and very fine and fine silt and very fine and fine sand, reflected in the high clr variances of clay and very fine and fine sand. The smallest variation in log ratios occurs between adjacent fractions (e.g., clay and very fine and fine silt, medium and coarse silt, coarse silt and very fine and fine sand), indicating more coupled behaviour. These patterns reflect how various particle sizes respond to environmental conditions. Finer fractions preferentially accumulate during interglacial pedogenesis, whereas coarser fractions increase under high-energy glacial wind regimes (Schulte and Lehmkuhl, 2018; Schwahn et al., 2023).

The compositional form and covariance biplots (Fig. 5) show that the first principal component (PC1) explains 90% of the dataset's variability, whereas PC2 explains only 7%. Clay and very fine and fine sand are well represented in the biplot (Fig. 5a). The samples show a wide range along PC1 but minimal dispersion along PC2, consistent with the explained variability of PC1. As shown in Figure 5b, negative PC1 scores correspond to compositions dominated by coarse silt and very fine and fine sand, whereas positive scores are dominated by clay and very fine and fine silt. This inverse relationship indicates that increases in fine fractions are coupled with decreases in coarse fractions, and vice versa.

PC1 can therefore be interpreted as a wind-energy gradient. Negative scores reflect high-energy glacial conditions favouring coarse particle transport, whereas positive scores reflect lower-energy interglacial or pedogenic conditions associated with finer fractions. This gradient closely parallels the EM4-EM1 contrast of Wacha et al. (2021) and provides a statistically robust axis for comparing grain-size variability with independent proxies. The medium silt fraction is associated with negative PC2 scores, suggesting deposition under moderate transport conditions. However, the low percentage of variability explained by PC2 indicates that such conditions represent a minor component of overall variability.

The transition from OIS 6 to OIS 5 (Fig. 7a) is defined by a slope lying (including confidence ellipses) on the 44-200 µm/16-44 µm isoproportion line, indicating a constant ratio between these fractions. During this transition, very fine and fine sand and coarse silt fractions decreased by approximately 3% each relative to clay, which increased by approximately 6% (Table 3). The transition is thus primarily expressed as an enrichment in clay. This reflects a shift from colder to warmer climatic conditions and an enhancement of pedogenesis. The constant ratio between the particle sizes of 44-200 µm and 16-44 µm suggests a steady weakening of the winds. Marković et al. (2015) describe this as a transition from periglacial loess accumulation to the development of forest-steppe or steppe soils in the Danube Basin.

The slope that describes the transition from OIS 5 to OIS 4 lies between the 44-200 µm/16-44 µm and 16-44 µm/<16 µm isoproportion lines (Fig. 7b). There was a relatively small change in the 16-44 µm fraction, approximately 2.5% (Table 3), whereas the relationship between the <16 µm and 44-200 µm fractions changed substantially in favour of the <44-200 µm fraction. The fraction <16 µm decreased by approximately 18.5%, and the fraction 44-200 µm increased by approximately 16%. This indicates a dominant shift in the ratio between fine and coarse fractions, with the intermediate fraction remaining stable. The pattern reflects strengthening winds capable of transporting coarser particles, consistent with colder, drier glacial conditions. This interpretation is supported by reduced pedogenic indicators reported by Wacha et al. (2021) and the regional environmental shift described by Marković et al. (2015).

Similar to the transition between the OIS 5 and OIS 4 stages, the slope for the transition from OIS 4 to OIS 3 lies between 44-200 µm/16-44 µm, and 16-44 µm/<16 µm isoproportion lines (Fig. 7c). The relationships among the fractions can be described by a relatively small decrease in the 16-44 µm fraction by approximately 2%, a substantial increase in the <16 µm fraction by approximately 12% and a decrease in the 44-200 µm fraction by approximately 10% (Table 3). Thus, the dominant indicator of the transition from OIS 4 to OIS 3 is the change in the ratio between the <16 µm and 44-200 µm fractions, whereas the 16-44 µm fraction remains roughly constant. Although less pronounced than the transition from OIS 5 to OIS 4, this shift indicates a move toward warmer and wetter conditions. The stability of the intermediate fraction again highlights its limited sensitivity to climatic forcing, while the opposing behaviour of fine and coarse fractions reflects changes in wind strength and pedogenesis. Wacha et al. (2021) interpret OIS 3 as a phase of weakened pedogenesis within generally glacial conditions.

The slope that describes the transition from OIS 3 to OIS 2 lies (including confidence ellipses) between the 16-44 µm/44-200 µm and 44-200 µm/<16 µm isoproportion lines (Fig. 7d). In general, the ratio between the <16 µm and 16-44 µm fractions changed, whereas the 44-200 µm fraction exhibited only a small change (a decrease of approximately 1.5%). Fractions 16-44 µm increased by approximately 4.5%, and fractions <16 µm decreased by approximately 6% (Table 3). Therefore, the transition between OIS 3 and OIS 2 dominantly influenced the change in the ratio between the 16-44 µm and <16 µm fractions, whereas the 44-200 µm fraction remained roughly constant. The stability of the 44-200 µm particles suggests that the threshold wind energy required for sand transport did not change substantially. Instead, the shift reflects a modest increase in wind strength sufficient to reduce fine fractions but not to significantly enhance sand transport. This interpretation is consistent with the already cold and dry conditions of OIS 3, which limited the magnitude of further environmental change into OIS 2 (Marković et al., 2015). Wacha et al. (2021) further note that this transition is often expressed more by reduced weathering than by pronounced grain-size coarsening. In addition, Újvári et al. (2016) emphasise that sediment supply, controlled by fluvial systems, can modulate or even override wind-energy signals in loess grain-size records.

The interpretation of grain-size transitions must, however, account for the interplay of aeolian deposition, pedogenesis, and potential post-depositional modification, as these processes are difficult to fully disentangle in loess-paleosol sequences (Aquino et al., 2024; Labahn et al., 2026). The clay enrichment observed during OIS 5 is consistent not only with reduced wind energy but also with pedogenesis. Wacha et al. (2021) documented increased smectite content in paleosol horizons at Zmajevac, which they interpreted in part as a product of in situ weathering of micas and feldspars under warmer, more humid interglacial conditions. Thus, an increase in the <16 µm fraction during OIS 5 likely reflects both a reduction in coarse aeolian input and an increase in pedogenic fine material, and these two signals cannot be fully separated with grain-size data alone (Vandenberghe et al., 2018).

Finally, the stratigraphic record at Zmajevac may include additional features such as weak paleosols. While these features are supported by the colorimetric data in the supplementary material of Wacha et al. (2021) and may introduce local variability, the high-resolution nature of the grain-size dataset ensures that these minor fluctuations do not significantly bias the statistical centres used for modeling. Consequently, the calculated slopes and isoproportion transitions reflect the primary paleoenvironmental shifts between OIS units rather than localised pedogenic overprinting.

5.2. Stable isotopes

The malacological assemblage of the Zmajevac LPS was previously characterised by Molnár et al. (2010) and Banak et al. (2012), with subsequent stable isotope analysis reported by Banak et al. (2016). Isotopic measurements focused on the $\delta^{18}O$ and $\delta^{13}C$ values obtained primarily from *H. striata* shells, supplemented by a minor component of *Arianta arbustorum* (Linnaeus, 1758).

Contemporary European terrestrial gastropods exhibit activity within the 10-27 °C temperature range, entering hibernation or dormancy below 10°C (Heller, 1990; Thompson and Cheny, 1996). Shell formation occurs predominantly during the active growing season, which spans 160-210 days from spring through autumn (Frömming, 1954; Ant, 1963), thereby recording average growing season (AGS) temperatures. Shell calcification is enhanced during and following precipitation events (Ward and Slotow, 1992), establishing a direct isotopic link between meteoric water and shell carbonate $\delta^{18}O$ values. These modern analogues provide the basis for interpreting fossil gastropod isotopic signatures.

The results of this study differ systematically from those of earlier Zmajevac LPS investigations (Banak et al., 2016) (Fig. 8). While the average $\delta^{13}C$ values remain consistent (this study: -7.19‰; Banak et al. (2016): -7.74‰), indicating that the stable vegetation composition is dominated by cold steppe grasslands, the $\delta^{18}O$ values are notably more positive (this study: -3.06‰; Banak et al. (2016): -4.20‰). $\delta^{13}C$ values are directly linked to snail diet and indicate whether the diet is based on C3 or C4 plants. C3 plants (e.g., most trees, shrubs, and cool-season grasses adapted to wet climates) present highly negative $\delta^{13}C$ values, typically ranging from -24‰ to -32‰ (average ~-27‰; Hatte et al., 2013). C4 plants (e.g., many warm-season grasses such as maize, sorghum, and plants adapted to arid, high-light environments) present relatively low $\delta^{13}C$ values, typically ranging from -9‰ to -17‰ (average ~-13‰; Hatte et al., 2013). The $\delta^{13}C$ values measured in our gastropod shells (ranging from -8.29‰ to -5.88‰, Table 5) must be understood in the context of 5‰ to 14‰ enrichment between diet (plant tissue) and shell carbonate (Stott, 2002). Therefore, our average shell values of ~-7.2‰ correspond to a diet with a $\delta^{13}C$ value of approximately -12‰ to -21‰. This value falls between those of the C3 and C4 vegetation biomes.

The most negative $\delta^{18}O$ values (-4.78‰ average) from Banak et al. (2016) occurred at the oldest stratigraphic levels, suggesting that our study interval represents a younger, potentially warmer period. When equivalent stratigraphic positions are compared, the average $\delta^{18}O$ values converge (-3.84‰ vs -3.06‰) (Fig. 8). The most positive $\delta^{18}O$ values (indicating warmer or drier conditions) are consistently associated with paleosol horizons. For example, the sample from the OIS 5 paleosol at 910 cm depth yields a value of -2.34‰, the most positive value in the entire sequence (Table 5). In contrast, the most negative $\delta^{18}O$ values (indicating cooler conditions) are found within the loess horizons, such as the -4.02‰ value at 170 cm depth (OIS 2/3). This pattern can be interpreted through the lens of pedogenesis. Paleosols represent periods of landscape stability, warmer temperatures, and enhanced biological activity. During these interglacial/interstadial periods, gastropods would have been active in a warmer climate and potentially incorporated more ^{18}O-enriched water derived from soil moisture that had undergone increased evaporation. Conversely, loess deposition dominated during colder, glacial periods characterised by greater wind energy and reduced evapotranspiration, leading to more negative $\delta^{18}O$ values recorded in the shells. This evaporative enrichment interpretation is further strengthened by the macro-regional malacological overviews of Molnár et al. (2020) and Banak et al. (2020). They demonstrated that along the Danube corridor of the southern Pannonian Basin, isotopic shifts in *Helicopsis striata* frequently exhibit a strong environmental sensitivity to local aridity and open-canopy exposure. The distinct enrichment trend we observed in paleosols confirms that warm-season evaporation of topsoil moisture significantly boosted absolute $\delta^{18}O$ shell values during periods of landscape stability. This effect occurred independently of changes in the base isotopic signature of regional meteoric precipitation.

Converting relative temperature changes to absolute values requires comparison with modern precipitation isotope data. Gastropod shell $\delta^{18}O$ values are typically enriched by ~5‰ relative to equilibrium with meteoric water (Lecolle, 1985; Goodfriend and Magaritz, 1987). The mean shell $\delta^{18}O$ value from this research of -3.06‰ therefore corresponds to a paleo-precipitation $\delta^{18}O$ of approximately -8.06‰.

To determine the paleotemperature from these data, the average $\delta^{18}O$ value of recent rainwater from the closest possible area must be known. The mean annual $\delta^{18}O$ value of rainwater during the AGS period (May, June, July, August, September) in the village of Sikirevci, located approximately 50 km south of Zmajevac, is -6.46 ‰ (Hunjak, 2015). The mean temperature during the AGS period for Sikirevci is +21.7 °C (Hunjak, 2015). If the aforementioned data are incorporated and compared with those of this study, it is clear that the average intermediary $\delta^{18}O$ value of paleo rainwater is on average 1.6‰ more negative than the modern value of rainwater from the village of Sikirevci. Estimated AGS temperatures and dominant vegetation character differed across stages. OIS 5 recorded the least negative mean $\delta^{18}O$ (−2.71‰), corresponding to an estimated AGS cooling of 3.6-5.0 °C relative to the modern value of +21.7 °C, and a mean $\delta^{13}C$ of −7.11‰, consistent with mixed vegetation broadly intermediate between C3 and C4 sources but dominated by C3 plants, suggesting a relatively humid forest-steppe environment. OIS 6 was isotopically similar (mean $\delta^{18}O$ −2.80‰, mean $\delta^{13}C$ −7.12‰), implying cooling of 3.8-5.4°C and a comparable mixed C3-dominated vegetation signal. OIS 4 shows a mean $\delta^{18}O$ of −3.02‰ (cooling of 4.5-6.2°C) and the most negative mean $\delta^{13}C$ in the record (−7.62‰), pointing to the densest C3 vegetation cover of any stage, consistent with persistent woodland elements under otherwise glacial conditions. OIS 2 has the most negative mean $\delta^{18}O$ (−3.20‰), indicating the greatest cooling of 5.0-7.0 °C relative to modern AGS, while the mean $\delta^{13}C$ of −7.26‰ reflects predominantly C3 vegetation. OIS 3 is represented by a single isotope sample (640 cm, $\delta^{18}O$ −3.27‰, $\delta^{13}C$ −5.88‰), which yields a cooling estimate of 5.2-7.2 °C but an anomalously enriched $\delta^{13}C$, suggesting stressed or C4-influenced vegetation - an outlier discussed in section 5.4 in the context of proxy decoupling. Across all stages, paleotemperatures were consistently below the modern AGS value, with OIS 5 being the warmest and OIS 2 the coldest interval recorded in the sequence.

The temperature reconstructions from this research align well with those from other Pannonian Basin studies. Kovacs et al. (2012) estimated mean annual temperatures of 6.2-11.2 °C for OIS 2 in the Czech Republic and Slovakia based on mammoth tooth enamel oxygen isotopes, yielding a temperature range of ~5 °C, which is comparable to the formula described in Balakrishnan et al. (2005). The significantly lower Central European temperatures (4.5 °C) derived from noble gas thermometry (Corcho Alvarado et al., 2011) likely reflect the more southerly position of the Zmajevac site within the Pannonian Basin.

5.3. Palynology

Despite the low abundance of palynomorphs, several inferences can be made. The sample from penultimate glacial loess deposited during OIS 6 contains sporomorphs predominantly derived from non-arboreal plants, indicating a steppe environment, as confirmed by palynofacies dominated by charcoals.

The sample from the strongly developed brown paleosol of the last interglacial contains a higher proportion of arboreal plant sporomorphs, suggesting the presence of a deciduous forest. However, the palynofacies remain dominated by phytoclasts from non-arboreal plants (herbs). This discrepancy likely reflects differences in transport and depositional processes, whereby arboreal sporomorphs, particularly *Pinus* spp. pollen, can be dispersed over variable distances, whereas non-arboreal sporomorphs are predominantly deposited in situ.

Palynofacies from samples attributed to the Lower Pleniglacial (OIS 4) are again dominated by charcoal, similar to those from OIS 6. Nevertheless, these samples contain a higher abundance of sporomorphs, including both arboreal and non-arboreal taxa, indicating the persistence of deciduous forest refugia. In the oldest OIS 4 sample, reworked dinoflagellate cysts originating from Miocene deposits were identified, pointing to erosion of these sediments, which are present as surface outcrops in the central and western parts of Bansko brdo hill (Hećimović, 1991; Pikija et al., 1991; see Fig. 2). In the middle part of this interval, the presence of the alga *Pediastrum* sp. suggests a freshwater environment (e.g., a pond). In the youngest OIS 4 sample, spores of the moss genus *Sphagnum* sp. were recorded, indicating wet, acidic, and humid conditions typical of peat bog environments associated with the boreal floral element. The late-glacial temperature increase may have promoted hydrological changes and the development of localised wet habitats. *Sphagnum* sp. thrives in waterlogged, stagnant, nutrient-poor conditions, often in acidic, sandy substrates.

In the sample from the Middle Pleniglacial (OIS 3), the palynofacies is dominated by charcoal, and *Sphagnum* sp. is again present, suggesting the persistence of similar wet environments such as peat bogs. This interpretation is supported by the occurrence of the alga *Mougeotia* sp., which inhabits a range of aquatic settings, including low-pH waters.

Samples from the Upper Pleniglacial (OIS 2) show some differences, with a higher proportion of sporomorphs from drought-tolerant plants, indicating a shift toward more arid conditions and steppe-like vegetation.

Notably, deciduous tree pollen occurs throughout the loess profile, suggesting the continuous presence of nearby refugial areas during glacial periods at the Zmajevac locality. However, better-preserved palynomorphs are required to confirm these paleoenvironmental interpretations with greater confidence.

5.4. Relationships between grain size distributions, isotope values and pollen

The multi-proxy record from Zmajevac allows the three independent datasets, grain-size, stable isotopes ($\delta^{18}O$ and $\delta^{13}C$), and pollen, to be compared across the OIS 2-6 sequence (Fig. 9). Where the proxies converge, they provide robust paleoenvironmental reconstructions, where they diverge, they reveal either time lags between environmental subsystems, local ecological overprinting, or taphonomic complexity.

All identified arboreal taxa (*Pinus* spp., *Picea* sp., *Quercus* sp., *Ulmus* sp., *Betula* sp., *Corylus* sp., *Carpinus* sp., *Ostrya* sp., *Tilia* sp., *Fagus* sp., *Alnus* sp., *Juglans* sp., *Fraxinus* sp., *Populus* sp., *Pterocarya* sp., *Taxus* sp.) are assigned to the C3 type. The majority of non-arboreal taxa, including *Ambrosia* sp., *Artemisia* sp., *Typha* sp., *Plantago* sp., *Xanthium* sp., members of Ericaceae, Caryophyllaceae, and Asteraceae (including the tribe Cichorieae), as well as ferns (*Polypodium* sp., *Pteridium* sp.) and mosses (*Sphagnum* sp.), are also C3 taxa. In contrast, Poaceae, Cyperaceae, and Amaranthaceae comprise both C3 and C4 taxa. Consequently, assignment to a specific photosynthetic type within these families cannot be resolved without species-level identification, which is not possible at the present level of preservation.

5.4.1. Transitions between OIS stages

The character of OIS transitions, as expressed by the PC1 signal, varies markedly across the profile (Fig. 9). The OIS 6/OIS 5 transition is gradual, i.e., PC1 values remain consistently positive on both sides of the boundary, indicating that relatively weak aeolian conditions and pedogenic processes were already established in the uppermost OIS 6 and continued through OIS 5 without an abrupt shift. The OIS 5/OIS 4 transition is the most pronounced in the record, indicating rapid intensification of aeolian activity. However, as noted by Poch et al. (2024), particle sorting and redoximorphic features in the upper part of OIS 5 indicate fluvial or sheet-wash reworking, meaning the apparent abruptness may partly reflect alluvial erosion of younger OIS 5 sediments rather than a purely climatic signal. The OIS 4/OIS 3 transition is also well expressed. PC1 rises progressively from its most negative values toward near-zero and slightly positive values at the boundary, reflecting a gradual weakening of winds. The OIS 3/OIS 2 transition is the least distinct in the PC1 record, with values hovering near zero on both sides of the boundary, consistent with the previous interpretation that this transition was dominated by a modest change in fine fractions rather than pronounced grain-size coarsening.

The $\delta^{18}O$ values broadly follow these trends, with the most positive values recorded in OIS 5 and progressively more negative values through OIS 4 into OIS 2, though the shifts are moderate rather than abrupt. The pollen signal, while sparse, is consistently dominated by non-arboreal plants (NAP) throughout the profile, with arboreal plants (AP) percentages ranging from only 9% (OIS 2) to 64% (OIS 5), reflecting the generally open, steppe-dominated character of the investigated locality across all recorded stages.

5.4.2. Coupling

The clearest coupling among all three proxies occurs within OIS 5 (891-1001 cm; Fig. 9). PC1 reaches its highest mean value in the entire record, indicating the weakest aeolian activity and most intense pedogenesis. Simultaneously, $\delta^{18}O$ reaches its least negative value at 910 cm (−2.34‰), the most positive in the entire sequence, recording the warmest or driest growing season signal. $\delta^{13}C$ at the same depth (−8.24‰) is among the most negative values in the record, consistent with dense C3 vegetation and high stomatal conductance under humid conditions, as modelled by Farquhar et al. (1982, 1989) and discussed in the context of loess sequences by Kohn (2010). Pollen sample Zm-9 (950-960 cm) records the highest AP quantity in the profile, with thermophilous taxa including *Quercus* sp., *Ulmus* sp., *Fraxinus* sp., and *Pterocarya* sp., although NAP taxa (*Ambrosia* sp., Amaranthaceae, Poaceae) remain present, indicating a forest-steppe environment rather than closed woodland. All three proxies thus converge on a warm, relatively humid interglacial environment with active pedogenesis and mixed woodland cover, providing the most robust paleoenvironmental signal in the record. The two deeper OIS 5 samples (950-980 cm) show a divergence between $\delta^{18}O$, which shifts toward more negative values, and $\delta^{13}C$, which trends toward less negative values. These values may reflect a lag attributable to slower vegetation reestablishment and taphonomic constraints on *Helicopsis striata* population recovery, with both the OIS 4 and OIS 5 signals potentially still capturing the weakening influence of the OIS 5 interglacial optimum (Lecolle, 1985; Goodfriend and Magaritz, 1987; Stott, 2002).

The upper part of OIS 2 (100-250 cm) shows a different type of coupling (Fig. 9). PC1 becomes slightly negative, indicating increased wind energy, $\delta^{18}O$ reaches its most negative value in the profile (−4.02‰ at 170 cm), and pollen from Zm-1 (100-110 cm) records the lowest AP quantity in the entire sequence, with NAP overwhelmingly dominant. These three signals consistently point to cold, dry, wind-dominated steppe conditions, a fully glacial environment. This constitutes the clearest cold-end coupling in the record.

5.4.3. Decoupling

The lower OIS 2 interval (350-500 cm, Zm-3 and Zm-4) shows a partial decoupling between pollen and PC1 (Fig. 9). PC1 values are near zero, suggesting limited aeolian activity, yet pollen records are among the highest arboreal values within OIS 2, alongside persistent NAP taxa. The isotopes at this interval ($\delta^{18}O$ −2.76 to −2.98‰, $\delta^{13}C$ −7.38 to −7.89‰) do not show a particularly warm or moist signal. This combination may reflect the persistence of local forest refugia in sheltered valleys during a phase of relatively low wind intensity within OIS 2, without a corresponding broad regional climate amelioration, consistent with the refugial woodland interpretation previously discussed by Molnár et al. (2021) and with the canopy effect explanation for the relatively negative $\delta^{13}C$ values noted by van der Merwe and Medina (1991) and Drucker et al. (2008).

The pronounced decoupling involves the pollen signal relative to grain size and isotopes in the OIS 3 interval (640-650 cm, Zm-5; Fig. 9). PC1 values are slightly positive throughout OIS 3, indicating weak winds and near-pedogenic conditions, which might suggest mild or transitional environmental conditions. $\delta^{13}C$ at 640 cm (−5.88‰) is the least negative value in the entire profile, pointing to open, C4-influenced or stressed vegetation. Yet pollen from Zm-5 (640-650 cm) shows AP as dominant taxa. This divergence may reflect the different spatial scales captured by each proxy, like grain-size records regional depositional energy, pollen integrates regional vegetation from varying source distances, and $\delta^{13}C$ records the diet of snails living in a specific microhabitat-refugia. The enriched $\delta^{13}C$ at 640 cm may alternatively reflect a taphonomic overprint or a brief episode of moisture stress affecting vegetation carbon discrimination without a corresponding change in regional wind regime (van der Merwe and Medina, 1991; Drucker et al., 2008).

Next decoupling occurs within OIS 4, between 700 cm and 810 cm (Fig. 9). $\delta^{13}C$ shifts to −8.29‰ and $\delta^{18}O$ shifts to −2.71‰ at 810 cm, suggesting progressively denser C3 vegetation and higher temperatures. In addition, pollen signal points to dominating AP. At the same time, PC1 simultaneously becomes more negative, indicating strengthening winds. Although *Pinus* spp. pollen (included within AP) can be dispersed over long distances due to the presence of air sacs, thus skewing the AP/NAP ratio in sample Zm-7 (800-810 cm), these anomalies may reflect a brief interstadial episode. However, it should be noted that Obreht et al. (2019) highlighted those proxies like grain-size often indicate cold, high-energy glacial conditions, whereas biological and chemical indicators in the Pannonian Basin frequently suggest warmer summer temperatures or increased moisture during the same intervals. They noted that while aeolian dynamics point to intensified glacials, malacological data can indicate mean July temperatures as high as 21°C (Marković et al., 2007), a contrast that mirrors the decoupling of wind-energy (PC1) and vegetation signals observed here.

OIS 6 (1001-1235 cm) presents a further and somewhat unexpected decoupling (Fig. 9). Despite representing a glacial stage, PC1 throughout OIS 6 is consistently positive, indicating conditions more reminiscent of weak aeolian activity than full glacial loess accumulation. $\delta^{13}C$ values (~−7.1‰) and $\delta^{18}O$ values (~−2.8‰) are moderate, and pollen from Zm-10 (1040-1050 cm) shows AP with *Pinus* spp., *Quercus* sp., *Ulmus* sp., *Corylus* sp., and *Pterocarya* sp. as dominant over NAP taxa (*Ambrosia* sp., *Artemisia* sp., *Xanthium* sp.), indicating a forest-steppe rather than full steppe. The pollen AP quantity at Zm-10, though modest in absolute terms, is higher than most OIS 2 samples and consistent with the relatively weak wind signal, partially suggesting a less severe glacial environment at the bottom of the section.